Measuring the Effects of Heat Lamp Exposure and Bottom Bleed Reduction in Binder-jetted Parts

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Abstract:

Binder jetting is an additive manufacturing (3D printing) process used to manufacture parts from a powder.

In binder jetting, a part is first modeled using a computer-aided design (CAD) software, as shown in Figure 1. It is then prepared for printing with a software that splits it into many layers. Powder is laid down during printing in the build envelope with the powder spreading tool, as shown in Figure 2. Binder is then released from the print-head into each layer. A heat lamp may be used to dry each layer before a new layer is added, and parts printed with bottom bleed reduction have less binder added to lower layers to prevent them from becoming distorted from having too much.

This research investigates how the surface roughness of a binder-jetted part can be changed by modifying the printing parameters of heat lamp exposure and bottom bleed reduction. Both factors were found to affect the surface roughness of the final part.

The results gained from this experiment provide insight and comparison between the methods of bottom bleed reduction and heat lamp exposure. They can be used to form stronger parts with the binder-jetting additive manufacturing method.

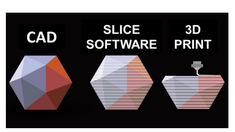

Figure 1: Schematic of Binder Jetting Process

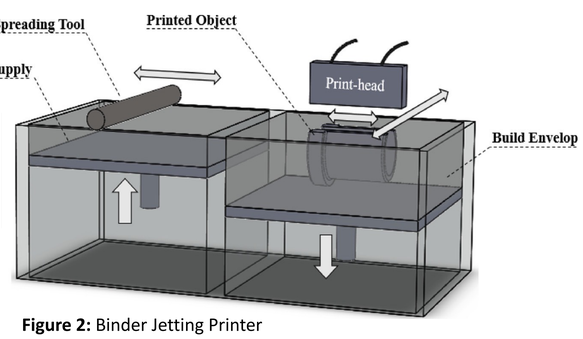

Figure 2: Binder Jetting Printer

Motivation:

One challenge that binder jetting currently faces is that binder-jetted parts can be rougher than parts made with other methods. This roughness can lead to parts with lower surface quality and can even cause pores in parts, which weaken them. This obviously presents a challenge for manufacturing the metal parts investigated in this research, whose high strength is often an essential design feature.

If binder jetting can be refined to create smoother, stronger parts, it could become a more effective and widely-used method for both prototyping and manufacturing of easily customizable parts.

Figure 3: Powder Spreader

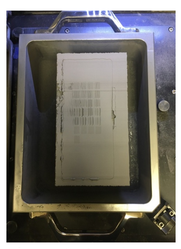

Figure 4: Powder Bed

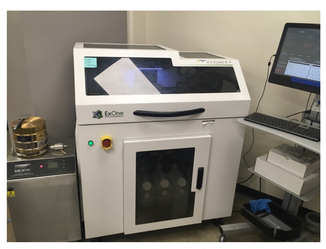

Figure 5: Binder Jetting Printer

Method:

Parts were printed using an ExOne Innovent+ Binder Jetting Printer. Figures 3, 4, and 5 provide a visual reference of an ExOne Innovent+ printer. The binder jetting process is also shown in Figures 1 and 2. The parts were then heat treated in an oven and measured with a microscope.

Method, *continued*:

The thickness of each layer in this experiment was set to 35 µm. Parts were printed 1, 2, 3, 5, and 8 layers thick. The powder used for each part was ExOne 316 stainless steel powder, and ExOne solvent binder was used In each part. These parameters attempt to follow those used by Colton, *et al.*

Bottom bleed reduction settings on the ExOne Innovent+ printer vary between zero and 75%/50%. 75%/50% means 75% reduction in binder added to the bottom layer and 50% reduction in the second layer. The bottom bleed reductions used in this experiment were 0%/0% for the "No Bleed Reduction" groups and 50%/50% for the "Bleed Reduction" groups. The heat lamp exposure times used in this experiment were 1 second, which is the minimum time allowed by the machine, and a 30 seconds, which like 50%/50% bleed reduction was chosen as a typical length of heat lamp exposure time.

The microscope used to examine these parts before removal from the powder bed was a Keyence VHX-7000 microscope, which has a feature for measuring a part's surface roughness. Twenty samples were measured of each part to investigate the variance in results.

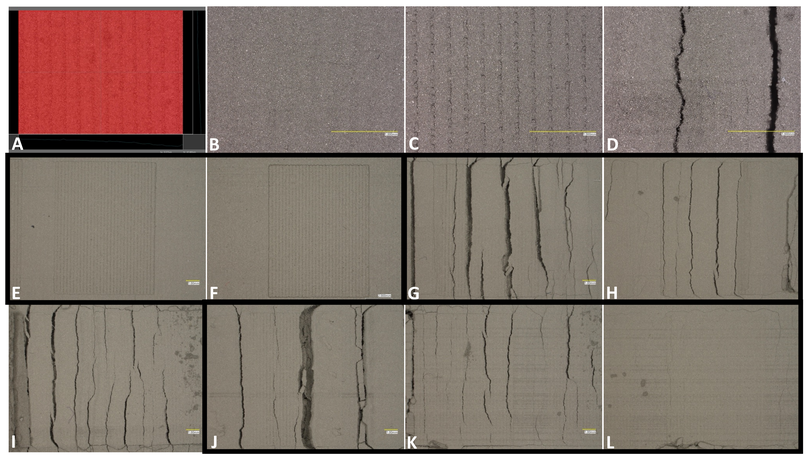

Figure 6: Images and Roughness Measurements Taken with Keyence VHX-7000 Microscope
Figure A shows a screenshot of the interface used to collect 20 measurements of each thickness.
Figures B, C, and D show parts at 100× magnification through the microscope, the magnification used to take measurements. Figures E, F, G, H, I, J, K, and L were taken at 20× magnification.
Figure E, a part printed without heat lamp exposure or bleed reduction, has a very similar appearance to the part in Figure F, printed with only bottom bleed reduction.
Figures G and H, which are 3-layer parts, display greater differences in roughness. The part in Figure G was printed without heat lamp exposure or bleed reduction, and Figure H had bleed reduction.
Figure I shows a 5-layer part printed without bottom bleed reduction. It is smoother than the part shown in Figure G, demonstrating how thicker parts printed without bottom bleed reduction have smoother surfaces.
Figures J and K show 8-layer parts printed without heat lamp exposure or bleed reduction. Differences between them explain some of the variance shown in Figure 7.
Figure L shows an 8-layer part printed with heat bottom bleed reduction. It is visibly smoother than the parts printed without bottom bleed reduction in Figures J and K.

Results:

Figure 6 shows close-up images of parts printed with different parameters and thicknesses. These images were all taken using the Keyence VHX-7000 microscope. The roughness of the parts was measured at 100× magnification because we determined that measurements taken at lower magnification were less accurate. Figure 6A shows how the surface measurements were taken with the microscope. Figures 6B, 6C, and 6D also show zoomed-in pictures. Figures 6E and 6F show that one-layer parts with different parameters still had similar surfaces, but Figures 6G and 6H show that there were visible differences after even 3 layers of binder. Figure 6I shows that surfaces without bottom bleed reduction began to smooth out after the part thickness passed 5 layers.

Results, *continued*:

Figures 6J, 6K, and 6L show the significant variance between parts made without bottom bleed reduction, compared to a relatively smooth part part printed using bleed reduction. Figure 7 shows a detailed view of the differences in the standard deviation of roughness of parts printed with bottom bleed reduction compared to those printed without it.

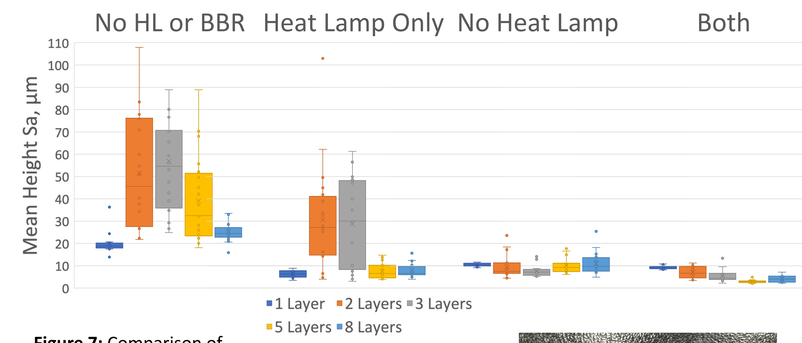

Figure 7: Comparison of Samples from Parts Printed with Different Parameters

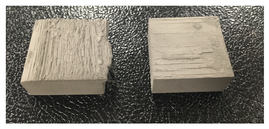

Figure 8: Parts From a Different Print Made Without and With Bottom Bleed Reduction

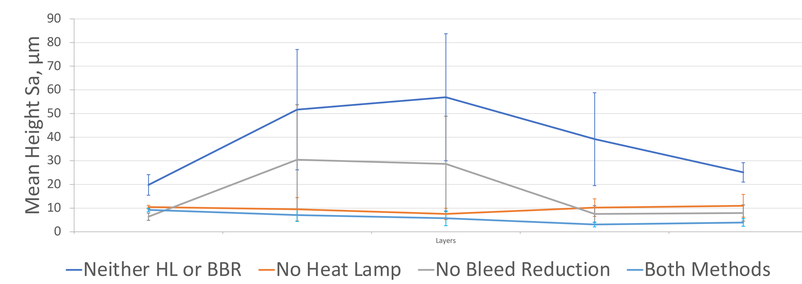

Figure 9: Superimposed Comparison of Parts Created Using all 4 Combinations of Parameters. Error bars represent sample standard deviations.

Conclusions:

This research found that both heat lamp exposure and bottom bleed reduction affect the final quality of binder jetted parts. Bottom bleed reduction appears to have a smaller effect in thicker parts, but Figure 8 shows that it is still very significant; it is only less visible from the surface. The use of both a heat lamp and bottom bleed reduction was found to be the best method for reducing surface roughness, with the line labeled "Both Methods" in Figure 9 showing that thicker parts even displayed lower surface roughness than thinner parts.

Between the individual parameters of bottom bleed reduction and heat lamp exposure, bottom bleed reduction also had the greatest general impact on final part surface roughness.

Interestingly, while omission of bottom bleed reduction seemed to have less effect at least on the top layers of thicker parts, omission of heat lamp exposure seemed to a greater impact on thicker parts than thinner parts. These results could be different with different layer thicknesses and droplet or line spacing, though.

Speed of printing was slower with heat lamp, but it yielded all-around best results. Because bottom bleed reduction is simply the application of less binder on lower layers, it does not have a significant effect on speed. Figure 5 shows that it does have a significant impact on the roughness of thicker parts.

References

Colton, T., Inkley, C., Berry, A., & Crane, N. B. (2021). Impact of inkjet printing parameters and environmental conditions on formation of 2D and 3D binder jetting geometries. *Journal of Manufacturing Processes*, *71*, 187–196. https://doi.org/10.1016/j.jmapro.2021.09.024

LiveRoom (2019, June 8). *3D Printing: The Internal Process and Development*. LiveRoom. https://liveroomlk.medium.com/3d-printing-the-internal-process-and-development-988ae9b24a6b

Ziaee, M., & Crane, N. B. (2019). Binder jetting: A review of process, materials, and methods. *Additive Manufacturing*, *28*, 781–801. https://doi.org/10.1016/j.addma.2019.05.031